

two complementary strategies. These are:

Flatpak for desktop applications:

Popular applications like Firefox, LibreOffice, and Visual Studio Code are installed via Flatpak, which sandboxes them from the system while maintaining full functionality.

```
flatpak install firefox
flatpak install libreoffice
```

Containers for development:

For developers and power users, ShaniOS includes Distrobox, creating containerised environments that feel like native Linux installations. You can run multiple Linux distributions simultaneously—Ubuntu for one project, Fedora for another—without conflicts.

```
# Create an Ubuntu development environment
```

```
distrobox create --image ubuntu:22.04
--name dev-env
```

```
# Install development tools inside the container
```

```
distrobox enter dev-env
sudo apt install nodejs npm python3
```

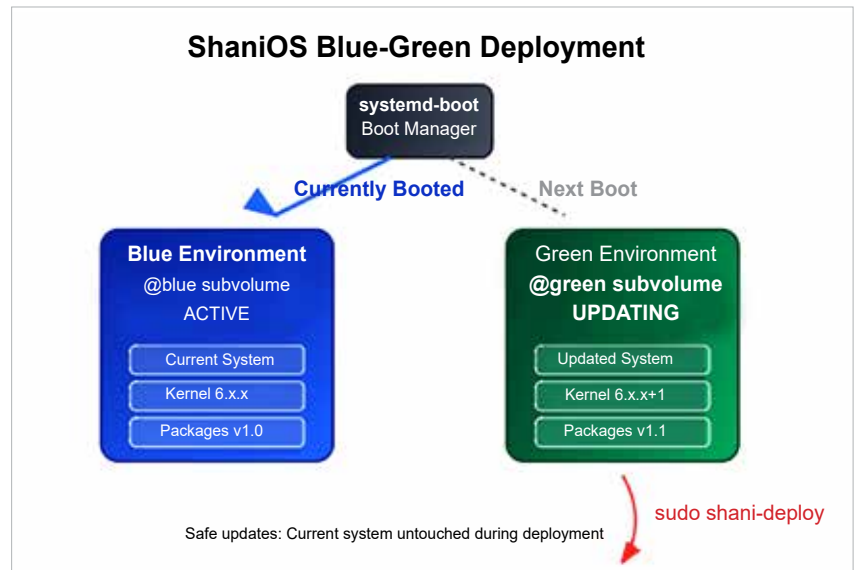


Figure 1: Blue-green deployment

Real-world benefits

For students and educators: Computer labs can update fearlessly without risking mid-semester disruptions. When updates cause issues, rolling back takes minutes instead of hours of reinstallation.

For developers: Different projects can utilise separate environments without interference. Your Python 2.7 legacy project won't conflict with your Python 3.11 development work.

For home users: Eliminates pre-update backup anxiety. Your system

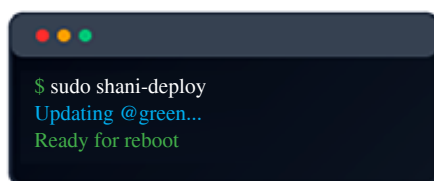
either updates successfully or remains exactly as it was—no intermediate broken states exist.

Security by design

ShaniOS incorporates enterprise-grade security features like:

- Full disk encryption with LUKS2
- Secure boot support with signed UKIs
- Immutable system files that malware cannot modify
- Application sandboxing through Flatpak containers

ShaniOS Update Process: Simple & Safe



Benefits:

- No system downtime
- Running system untouched
- Instant rollback capability
- User data always preserved

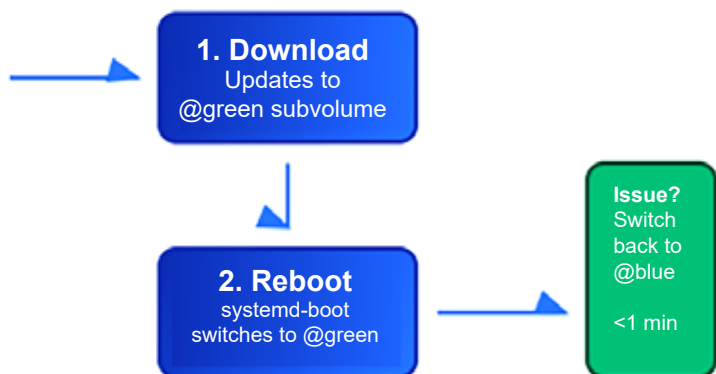


Figure 2: ShaniOS update process